

Hartshorne Chapter 1

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Section 1.1

Question 1.1

a)

$$A(Y) := k[x, y]/\langle y - x^2 \rangle = k[x][y]/\langle y - x^2 \rangle \simeq k[x][x^2] = k[x]. \quad \square$$

b)

Suppose that $A(Z) := k[x, y]/\langle xy - 1 \rangle \simeq k[t]$. Then note that $\overline{xy} = 1$, i.e. both $\overline{x}$ and $\overline{y}$ are units. Then the image of x, y under any such isomorphism must be in k as the units of $k[t]$ are exactly the elements of the base field. But then the image of the entire ring is k , and is thus not an isomorphism. $\square$

c)

Let $ax^2 + bxy + cy^2 + dx + ey + f \in k[x, y]$ be some arbitrary degree two polynomial. There are three cases.

1. $a = c = 0$. First send x to $x - e$, and extend everything else linearly to get an automorphism of $k[x, y]$. Observe that the x -term is cancelled out, such that we can assume $d = 0$ WLOG. Equally the automorphism given by $y \mapsto y - d$ allows us to assume $e = 0$. Hence we get that the polynomial is isomorphic to $bxy - f$ where b is necessarily nonzero as it must be a degree two polynomial and f is as well or else the polynomial is no longer irreducible. Normalize by taking $x \mapsto \frac{f}{b}x$ and factor out the common term f to get an isomorphism from the initial polynomial to $xy - 1$. Hence $A(W) \simeq A(Z)$ and we are done.
2. $a \neq 0, c = 0$. First using only the $a \neq 0$ assumption, the automorphisms $x \mapsto x - \frac{b}{2a}y$ eliminates b , and $x \mapsto x - \frac{d}{2a}$ eliminates d . Hence the initial polynomial is isomorphic to $ax^2 + cy^2 + ey + f$. Now normalize everything via $x \mapsto \frac{1}{\sqrt{a}}x, y \mapsto \frac{-1}{e}y$ to get $x^2 - y + f$. Finally via $y \mapsto y - f$ we get $x^2 - y$. Hence $A(W) \simeq A(Y)$ and we are done.
3. $a, c \neq 0$. By the result above we get $ax^2 + cy^2 + ey + f$. The isomorphism $y \mapsto y - \frac{e}{2c}$ eliminates e . Next normalize and scale everything via $x \mapsto \sqrt{\frac{-f}{a}}, y \mapsto \frac{-f\sqrt{-1}}{c}y$ and factor out $-f$ to get $x^2 - y^2 - 1 = (x + y)(x - y) - 1$. But then $k[x, y] = k[x + y, x - y]$, so it is isomorphic to $xy - 1$. Hence $A(W) \simeq A(Z)$ and we are done. $\square$

Question 1.2

Consider the quotient ring homomorphism $\pi : k[x, y, z] \rightarrow k[x, y, z]/\langle y - x^2, z - x^3 \rangle$. Then $f(x, y, z) \in \ker \pi \iff \pi(f(x, y, z)) = f(\pi(x), \pi(y), \pi(z)) = f(x, x^2, x^3) = 0 \iff f \in I(Y)$. Hence $I(Y) = \langle y - x^2, z - x^3 \rangle$, and $A(Y) = k[x, y, z]/\langle y - x^2, z - x^3 \rangle \simeq k[x]$. Note that $k[x]$ has dimension 1, so Y is a dimension 1 affine variety. $\square$

Question 1.3

To describe the algebraic set $Y \subset \mathbf{A}^3$, it suffices to find all points that vanish on the generating polynomials.

If (x_0, y_0, z_0) satisfy $zx - x$ then either $x_0 = 0$ or $z_0 = 1$ as k is a field.

If $x_0 = 0$, then $-y_0 z_0 = 0$ and thus $y_0 = 0$ or $z_0 = 0$ once again. These correspond to the lines $\langle 0, 0, t \rangle$ and $\langle 0, t, 0 \rangle$ where t varies over k . Moreover their ideals are $\langle x, y \rangle$ and $\langle x, z \rangle$ respectively, where both are prime ideals and hence correspond to irreducible components of Y .

If $z_0 = 1$ Then $x_0^2 = y_0$. This corresponds to the curve $\langle t, t^2, 1 \rangle$ with ideal $\langle x^2 - y, z - 1 \rangle$. Note that $k[x, y, z]/\langle x^2 - y, z - 1 \rangle \simeq k[x, x^2, 1] = k[x]$ so it is a prime ideal, and so this is the third and final irreducible component. $\square$

Question 1.4

We have that the Zariski topology $\mathbf{A}^1$ is exactly the cofinite topology.

But then notice that $(\mathbf{A}^1 \times (\mathbf{A}^1 \setminus \{y_0\})) \cup ((\mathbf{A}^1 \setminus \{x_0\}) \times \mathbf{A}^1) = (\mathbf{A}^1 \times \mathbf{A}^1) \setminus (x_0, y_0)$ such that $\mathbf{A}^1 \times \mathbf{A}^1$ is the discrete topology.

In particular, let (x_0, y_0) be some open point such that its complement is equal to $Z(I)$ for some $I \triangleleft k[x, y]$. But then if $f(x, y) \in I$ then f has the property such that $f(x_1, y) = 0$ for all $x_1 \neq x_0$, such that f is divisible by infinitely many polynomials of the form $(x - x_1)$.

Hence $f = 0$ necessarily, and $I = \{0\}$, which is a contradiction as $Z(\{0\}) = \mathbf{A}^2$. $\square$

Question 1.5

B is a finitely generated k -algebra with no nilpotent elements iff $B \simeq k[x_1, \dots, x_n]/\mathfrak{a}$ as k -algebras, where $\mathfrak{a} \triangleleft A$ such that $\sqrt{\mathfrak{a}} = \mathfrak{a}$.

But then by Hilbert's Nullstellensatz, the coordinate ring of every algebraic set in $\mathbf{A}^n$ is isomorphic to $A/\mathfrak{a}$, where $\mathfrak{a}$ is a radical ideal. Hence we get equivalence. $\square$

Question 1.6

Skipped.

Question 1.7

Skipped.

Question 1.8

Let $\mathfrak{p} := Z(Y)$ and $Z(H)$ be given by the irreducible polynomial f . Then note that $Z(Y \cap H) = \langle \mathfrak{p}, f \rangle$, such that the irreducible components of $Y \cap H$ correspond to minimal prime divisors of the ideal $\langle \mathfrak{p}, f \rangle$. As $Y \subsetneq H$, $\langle f \rangle \subsetneq \mathfrak{p}$. Hence by the correspondence theorem, such minimal prime divisors $\mathfrak{q}$ correspond to minimal prime divisors $\bar{\mathfrak{q}}$ of $\bar{f} \neq 0$ in $A(Y) \simeq k[x_1, \dots, x_n]/\mathfrak{p}$, so $\text{ht}(\bar{\mathfrak{q}}) \geq 1$. But then by Krull's Principal Ideal Theorem $\text{ht}(\bar{\mathfrak{q}}) \leq 1$, and $\{0\} \subsetneq \bar{\mathfrak{q}}$ such that $\text{ht}(\bar{\mathfrak{q}}) = 1$. Hence $\dim(\mathfrak{q}) = (n - r) + 1$, such that $\dim(Z(\mathfrak{q})) = n - ((n - r) + 1) = r - 1$. $\square$

Question 1.9

Straightforward application of Krull's height theorem. $\square$

Question 1.10

a)

It suffices to prove that if Y is a subspace of X such that Z is irreducible in Y , then it is irreducible in X .

Suppose that $Z = A \cup B$ in X where A and B are closed. Then $Z = (A \cap Y) \cup (B \cap Y)$ in Y , such that WLOG $B \cap Y = \emptyset$. But then $B \subseteq Y$ such that $Z \subseteq A$, and we are done. $\square$

b)

First, by (a) we have that $\dim X \geq \sup \dim U_i$.

Now let $Z_0 \subsetneq \cdots \subsetneq Z_n \subsetneq \dots$ be a chain of irreducible closed subsets in X . Next, notice if there exists U_1, U_2 in the open cover such that $Z_0 \cap U_1, Z_0 \cap U_2 \neq \emptyset$, $\{U_1^C \cap Z_0, U_2^C \cap Z_0\}$ is a closed cover for Z_0 such that it is not irreducible, which is a contradiction. Hence there exists a U_0 such that $Z_0 \subseteq U_0$.

Now, by 1.6, $Z_i \cap U_0$ is dense in Z_i for all i . Hence if there existed $Z_i \subsetneq Z_{i+1}$ such that $Z_i \cap U_0 = Z_{i+1} \cap U_0$, taking closures on both sides we observe that $Z_i = Z_{i+1}$, which is not possible. Hence $Z_0 \cap U_0 \subsetneq \cdots \subsetneq Z_n \cap U_0 \subsetneq \dots$ is a chain of irreducible closed subsets in U_0 , and $\dim X \leq \sup \dim U_i$. $\square$

c)

Consider the topology $\mathcal{T}$ on $[0, 2]$ where the open sets are given to be $\{\emptyset, [0, 1), [0, 2]\}$. Then $\dim(\mathcal{T}) = 1$ but $\dim([0, 1)) = 0$. $\square$

d)

Suppose that Y is a closed subset of a irreducible finite dimensional topological space X , such that $\dim Y = \dim X$.

Suppose that $Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_n$ is a maximal chain of irreducible closed sets. It suffices to prove that there exists no closed sets between Z_n and X . Hence, if it is nonempty, let $Z := \bigcap_{Z_n \subsetneq Z' \subseteq X} Z'$ be the intersection of all such closed sets. Then Z cannot be irreducible, such that $Z = A \cup B$. Then Z_n must be contained entirely with A or B as it is irreducible, such that WLOG $Z = A \subsetneq Z$ and we get a contradiction. $\square$

e)

Consider the topology on $[0, \infty)$ where the closed sets are given to be $\{\emptyset, [0, 1], [0, 2], \dots, [0, \infty)\}$. Then notice that the closed sets form a totally ordered chain, such that their finite union and arbitrary intersection is well-defined and also of the form $[0, n]$. But then every closed set is irreducible such that the topology has infinite dimension. It can also be seen that this topology is Noetherian as $[0, n]$ contains exactly n nonempty closed sets. $\square$

Question 1.11

Consider the map $\phi : \mathbf{A}^1 \rightarrow Y$ given by $t \mapsto (t^3, t^4, t^5)$. It is clearly surjective. Now suppose that $(t^3, t^4, t^5) = (t'^3, t'^4, t'^5)$. Then by considering the first coordiante, t and t' can only differ by a third root of unity. On the other hand by considering the third coordinate they can only differ by a fifth root of unity; hence $t = t'$ and ϕ is bijective.

Next, the Zariski topology on $\mathbf{A}^1$ is exactly the cofinite topology, such that the image of a proper closed set under ϕ must be a finite set. But then singletons are closed as they correspond to maximal ideals in $k[x, y, z]$, such that the finite collection of points is also closed. The map is closed.

Finally suppose that $S = Z(I) \cap Y$ is some proper closed set in Y . By definition, a point (t^3, t^4, t^5) is in S iff there exists an $f(x, y, z) \in I$ such that $f(t^3, t^4, t^5) = 0$. But then t must also be a root of the polynomial $f(x^3, x^4, x^5)$. The map is continuous, and defines a bijection. Thus $\dim(Y) = \dim(\mathbf{A}^1) = 1$. Hence we have that $\text{ht} I(Y) = \dim(k[x, y, z]) - \dim(Y) = 2$.

Now, by a similar proof as in 1.2, $A(Y) = k[x, y, z]/\langle x^5 - z^3, x^4 - y^3, y^5 - z^4 \rangle$.

Remark. how the hell do you prove it can't be generated by 2 elements.

Question 1.12

Let $f(x, y) := (x^2 - 1)^2 + y^2 \in \mathbb{R}[x, y]$. Then it is irreducible, but $Z(f) = \{(1, 0), (-1, 0)\}$ which is not irreducible. $\square$